

Smarter Cancer Staging: Using Machine Learning to Find What Stage a Cancer Is In



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Table of Contents

Note to Judges	4
Chapter 1. Executive Summary	5
Chapter 2. Introduction	7
2.1 Why Cancer Staging Matters for My Project	7
2.2 Why Current Staging Methods Are Limited.....	9
2.3 Why Gene Expression Is a Relevant Data Source	10
2.4 My Research Question and Aim	11
2.5 What This Project Adds Beyond Existing Research	12
Chapter 3. Literature Review	13
3.1 Existing Approaches to Cancer Staging	13
3.2 Gene Expression in Breast Cancer Progression	15
3.3 Machine Learning Models in Cancer Research	16
3.4 What Previous Studies Leave Unanswered	18
Chapter 4. Methodology	20
4.1 Overall Project Design	20
4.2 Dataset Selection and Structure.....	22
4.3 Feature Selection and Dimensionality Reduction	24
4.4 Model Selection and Training Process	25
4.5 Testing and Evaluation Strategy	27
Chapter 5. Results	28
5.1 Effect of Gene Panel Size on Model Performance	28
5.2 Behaviour on an Independent Dataset (SCAN-B)	30
5.3 ROC Curve Analysis Using Logistic Regression	32
5.4 ROC Curve Analysis Using XGBoost	33
5.5 Results Integrated Across Analyses.....	34
Chapter 6. Discussion	36
6.1 What My Results Actually Show	36
6.2 Why Performance Is Limited	38
6.3 Comparison with the Literature	40
6.4 Contribution of This Project	41
6.5 Handheld Demonstration Device and Computational Feasibility.....	42
6.6 Project Novelty and Practical Significance.....	43
Chapter 7. Conclusion	45
Chapter 8. References (APA)	47
Data Sources	47

Appendices	48
Appendix A: Final 50 Gene Panel	48
Appendix B : XGBoost Model Technical Specification	49
Task.....	49
Input feature space	49
Gene panel	49
Preprocessing	49
Model class.....	50
Class imbalance handling.....	50
Decision rule	50
Training configuration	50
Validation protocol.....	51
Failure modes / constraints	51
Appendix C: ESP32-S3 Demonstration Device	52
Purpose of the Embedded Implementation	52
Hardware as a Model Execution Target.....	52
Model Representation on the Device	53
Input Dimensionality and Gene Ordering.....	53
On-Device Normalisation Strategy	53
Inference Computation	54
Decision Thresholding.....	54
Output Interpretation	54
Scope and Limitations	55
Appendix D: Visuals	56
Appendix E: Reproducibility and Data Integrity Summary	58
Datasets Used	58
Data Access and Preprocessing	58
Feature Selection and Dimensionality Control	58
Model Training and Evaluation Safeguards	59
Randomness Control and Determinism	59

Note to Judges

This report is written to be accessible to readers without a specialist background in genomics or machine learning. Technical terms are explained at first use, acronyms are defined when introduced, and plain-language explanations are included where they aid clarity.

The main narrative of the project can be followed by reading:

- Chapter 1 (Executive Summary)
- Chapter 2 (Introduction)
- Chapter 5 (Results)
- Chapter 6 (Discussion)
- Chapter 7 (Conclusion)

Readers seeking detailed information on how the analysis was conducted may refer to Chapter 4 (Methodology), while Chapter 3 (Literature Review) provides the scientific context that motivated the project.

The report is structured to move from motivation, to evidence, and then to interpretation. The rationale for the project is set out in Sections 2.1–2.4, the modelling and validation choices are explained in Sections 4.1–4.5, and the results are interpreted and contextualised in Sections 6.1–6.6.

Figures and tables are designed to be interpretable on their own, with captions explaining both what is shown and why it is relevant. Performance results are presented with explicit justification, and the limits of the approach are discussed openly to avoid overstating what the project demonstrates.

Chapter 1. Executive Summary

Cancer stage is defined anatomically, yet this project shows that disease progression leaves a measurable footprint in gene expression without any staging information. The core contribution of this project is not higher accuracy, but isolation of the biological limit of gene expression-based staging. This project tests whether gene expression data alone contains usable information about cancer stage and whether this information could support staging decisions earlier in the diagnostic process, before scans and pathology are complete. Because anatomical information was removed by design, this analysis functions as an *upper-bound experiment*, where observed performance reflects a limit set by gene expression rather than by model choice.

This framing is intentionally stricter than most machine learning studies in cancer research. Many existing studies include anatomical or clinical variables, which makes it difficult to separate the biological contribution of gene expression alone.

Cancer stage directly affects hospital resource allocation, with later-stage cases prioritised for intervention. In current clinical practice, staging depends on imaging and pathology, which are time-consuming and resource intensive. Gene expression data is often available earlier in diagnosis, creating an opportunity to extract stage-related information before full anatomical assessment is complete. This is a difficult task because gene expression only measures molecular activity inside cells.

Imaging results, tumour size, lymph node involvement, and treatment history were excluded to avoid *circular reasoning* and to ensure that any predictive signal found had arisen from molecular biology and not from any existing staging methods. The TCGA Breast Invasive Carcinoma dataset was used, providing RNA-seq gene expression data (measurements of how active each gene is) and confirmed stage labels for approximately 1,200 patients. Expression values were standardised so that observed patterns did not arise from abnormal data processing.

Each sample initially contained measurements for approximately 20,000 genes. To manage this high dimensionality, the feature space was reduced to compact gene panels of 30 to 50 genes. This reduction transformed a research-scale dataset into a form that is computationally efficient and compatible with realistic laboratory workflows. Across multiple machine learning models, performance values were in the same ballpark, telling us that the results were not limited by model complexity but by biological limits.

Models achieved accuracy values of approximately 0.68 (68%) and ROC-AUC values between 0.61 and 0.66 (61% and 66%). ROC-AUC measures how well a model separates early-stage from late-stage cases across different thresholds. These results show the presence of a consistent stage-related signal in gene expression, with performance reaching a *plateau* as panel size increased. All models outperformed a *majority-class baseline*, confirming that predictions relied on structure in the data rather than simply predicting the most common class. When applied to an independent Swedish breast cancer cohort (SCAN-B) containing approximately 3,400 patients, model behaviour remained stable, indicating that the *extracted signal* transfers beyond a single dataset.

Once gene expression data is available, the system can flag samples whose molecular patterns align more closely with later-stage disease. This directly reflects the same signal observed during model evaluation and supports earlier prioritisation of imaging, pathology review, and multidisciplinary discussion within existing clinical workflows. This earlier prioritisation can reduce delays while leaving established staging criteria and clinical judgement unchanged.

Model inference was also tested on low-power embedded hardware (ESP-32), showing that computational requirements are minimal and does not limit an introduction into hospitals.

This work does not show that gene expression can replace anatomical staging. Instead, it shows that stage-related biological signal is already present early in the diagnostic timeline.

Chapter 2. Introduction

2.1 Why Cancer Staging Matters for My Project

Cancer stage was chosen as the focus of this project because it directly affects how cancer is treated. In hospitals, treatment decisions depend on whether a patient is classified as early or late stage, with later-stage cases normally prioritised for urgent intervention. For this reason, I focused on staging rather than *prognosis*, which predicts patient outcomes, or *subtype classification*, which groups cancers based on molecular features. While those areas are indeed important, staging links a bit more direct to immediate clinical decisions and was better suited to the aims of this project.

Gene expression data was used to test whether biological information could contribute to cancer staging. In standard clinical practice, staging relies on physical assessments such as imaging and biopsies to determine how far a cancer has spread. I aimed to test whether gene expression on its own still contains information related to cancer stage.

The *TCGA Breast Invasive Carcinoma dataset* was selected because it includes both gene expression data and confirmed cancer stage labels. It is also large enough to support analysis within a *student research project*. The dataset contains approximately 1,200 patients and measurements for around 20,000 genes per patient.

I used pre-processed expression values provided by TCGA rather than processing raw sequencing data. These values are produced using standard and widely accepted methods, reducing the chance that observed patterns arise from data handling.

Because gene expression data is high-dimensional, all ~20,000 genes were not used as model inputs. I tested smaller gene sets to reduce complexity and limit overfitting, where a model learns noise instead of general patterns. These tests showed that gene panels of approximately 30–50 genes provided the best balance between performance and stability. Reducing the number of genes kept the models simpler and focused on the

most relevant biological information. It also allowed the models to run on a small edge device for fast analysis, as discussed in Section 6.5.

Imaging data such as mammograms, MRI scans, and histopathology slides were not included, even though they are central to clinical staging.

Including imaging would have shifted the project toward image analysis rather than molecular data. By excluding imaging, I kept the analysis focused on a single question: whether gene expression alone contains information related to cancer stage.

Clinical variables such as tumour size, lymph node counts, and treatment history were also excluded. These variables are already used to define cancer stage, so including them would risk circular reasoning, where a model appears to predict stage using information that is part of the stage definition itself. Restricting inputs to gene expression ensured that any predictions were driven by biological differences rather than predefined clinical criteria.

Cancer stage was grouped into early and late categories. More detailed stage groupings exist, but finer distinctions are often less stable across patients and datasets. An early versus late split provided a more reliable classification task while still reflecting a difference that strongly influences treatment decisions.

While many previous machine learning studies include anatomical variables or shift focus toward prognosis prediction, this project deliberately isolates stage-related signal by strictly excluding staging inputs. This creates a more constrained and less common experimental setting.

2.2 Why Current Staging Methods Are Limited

Current cancer staging relies mainly on imaging and pathology. Imaging methods such as mammography, ultrasound, MRI, and CT scans require specialised equipment and trained staff. This makes them expensive and means access can vary depending on hospital resources, location, and waiting lists. As a result, full staging information is not always available quickly.

Biopsies and pathological assessment also have limits. A biopsy samples only a small part of a tumour, and cancers are biologically complex, with differences between regions of the same tumour. A small sample may therefore fail to capture the full disease. Repeated sampling is often impractical or unethical, so staging decisions are sometimes made using incomplete biological information.

Time delays add further limits. Imaging, biopsies, pathology review, and multidisciplinary discussions usually take place over weeks rather than days. Staging progresses step by step, and each step introduces additional waiting time. These delays can slow treatment planning and increase uncertainty for patients.

Because of these limits, gene expression data was examined as an additional source of information to support staging. I aimed to test how much stage-related information exists in gene expression alone and whether it could support faster decision-making when used alongside current clinical tools.

2.3 Why Gene Expression Is a Relevant Data Source

Gene expression was chosen as the main data source because it reflects what cells are actively doing at a given time. DNA sequence data remains mostly constant, whereas gene expression changes as cells respond to their environment. RNA sequencing measures how much RNA each gene produces, providing a snapshot of ongoing biological activity such as cell growth, metabolism, stress response, and signalling. This is relevant because cancer stage relates to progression and spread, which are changing processes rather than fixed genetic changes.

Gene expression was expected to change as cancer progresses because tumour cells face different biological pressures over time. Early-stage tumours are usually localised, while later-stage tumours must adapt to invasion, immune pressure, and survival in new environments.

Public gene expression datasets were deliberately used to keep this project transparent and repeatable. The TCGA Breast Invasive Carcinoma cohort served as the main dataset because it provides RNA-seq gene expression data alongside carefully checked clinical stage labels at a suitable scale for analysis. To test whether observed patterns were specific to TCGA, I also used the SCAN-B cohort. This independent Swedish breast cancer dataset contains approximately 3,400 patients, was generated using different laboratory methods, and represents a different population. Using both datasets allowed assessment of whether stage-related signals in gene expression persist across datasets rather than reflecting dataset-specific effects.

2.4 My Research Question and Aim

This project was structured around a single, clearly stated research question:

Can machine learning models distinguish between early-stage and late-stage breast cancer using gene expression data alone?

This question guided all major design choices, including dataset selection, model construction, and evaluation strategy. I kept the question deliberately narrow so that it could be tested directly and because the early versus late-stage distinction has clear consequences for clinical decision-making.

The task was framed as a binary classification problem. Cancer stages I to IIB were grouped as early-stage disease, while stages III to IV were grouped as late-stage disease. This grouping reflects a meaningful shift in disease progression and treatment approach. It also avoids the uncertainty and inconsistency that can arise when attempting to predict individual stage categories across different patients and datasets.

Framing the problem in this way reduced the risk that model behaviour was driven by unclear or unstable labels rather than biological signal. This kept the analysis focused, improved interpretability, and aligned the modelling task closely with the main aim of the project.

2.5 What This Project Adds Beyond Existing Research

Most existing research on cancer staging relies heavily on anatomical information such as tumour size, lymph node involvement, and imaging findings. When gene expression data is used, studies often focus on predicting prognosis or molecular subtype rather than cancer stage itself. Many studies also rely on very large gene signatures, test performance within a single dataset, or do not consider how results could fit into real hospital workflows.

This project differs from most existing work in three important ways.

First, it tests whether cancer stage leaves a detectable signal in gene expression without using any information that directly defines stage. Imaging data, tumour size, lymph node status, and treatment history were deliberately excluded. I made this choice to isolate how much stage-related information gene expression alone can provide and to ensure that results are not driven by clinical variables already used in staging.

Second, the project compresses this information into small gene panels of 30 to 50 genes. Using compact panels reduces complexity, improves stability, and makes the approach faster and more practical than using whole-transcriptome data. Small gene panels are also more suitable for routine laboratory workflows than very large gene signatures.

Third, the analysis explicitly tests whether results hold across datasets by applying the same gene panels and models to an independent international dataset generated using different laboratory methods. The focus is on consistent behaviour across cohorts rather than high performance within a single dataset.

When gene expression data is available, it can support earlier prioritisation of cases for imaging, pathology review, and further assessment. This fits within existing clinical workflows without replacing clinical judgement or established staging methods.

Chapter 3. Literature Review

3.1 Existing Approaches to Cancer Staging

Existing approaches to cancer staging were reviewed to understand what information they use, where their limits arise, and how those limits relate to the research question addressed in this project. The review focused on breast cancer because this project tests whether gene expression can capture information not fully described by anatomical and imaging-based methods. Reviewing these approaches helps explain why uncertainty can remain in staging decisions even when established clinical systems are used.

TNM staging was examined first. This system classifies cancer based on tumour size, lymph node involvement, and the presence of distant metastasis. It provides a widely used framework that allows clinicians to describe disease extent in a consistent and standard way. TNM staging is based almost entirely on anatomical observations. It describes where a cancer is and how far it has spread but does not measure how tumour cells behave at a biological level. As a result, tumours with similar TNM stages can progress at different rates or respond differently to treatment, because these differences are often driven by molecular processes not captured by anatomical staging.

Imaging-based staging methods were also reviewed, including mammography, ultrasound, MRI, CT, and PET scans. These tools are essential for visualising tumour extent and detecting possible metastases. Imaging mainly reflects large-scale structure and tissue-level features. It does not measure gene activity or biological pathways inside tumour cells and cannot resolve molecular differences between tumours that appear similar on scans. Important biological variation can therefore remain hidden when staging relies on imaging alone.

Molecular approaches to cancer staging, including gene expression analysis and other omics-based methods, were also considered. These approaches aim to capture biological processes involved in tumour

growth, invasion, and adaptation. Molecular data has been widely used for cancer subtyping, prognosis, and treatment response. However, many molecular studies are tested under controlled conditions or within a single dataset. Models trained in this way often perform poorly when applied to data generated using different laboratory methods or drawn from different patient populations, which limits how well results transfer to new settings (Vabalas et al., 2019; Pudova et al., 2025).

Across these approaches, clear limits emerge. Anatomical staging does not reflect biological behaviour, imaging does not measure molecular activity, and molecular data alone does not guarantee stable or transferable stage prediction. No single approach consistently links biological signal to reliable stage classification across diverse patients. This motivates careful evaluation, external testing, and cautious interpretation.

Together, these limits leave open a specific unresolved question: whether biological signal alone, isolated from all anatomical information, still contains usable information about cancer stage.

3.2 Gene Expression in Breast Cancer Progression

Gene expression reflects how tumour cells are behaving at the time a sample is collected. It does not encode tumour size, lymph node involvement, or the presence of metastasis. In this project, any ability to distinguish cancer stage therefore had to arise from biological activity alone rather than from anatomical information.

A clear separation between early-stage and late-stage disease is not expected from biology alone. Cancer progression involves gradual changes across many biological processes rather than a single clear shift. As a result, no individual genes separate stages cleanly. Instead, weak differences are distributed across many genes, meaning that any stage-related signal is expected to be subtle and spread rather than concentrated in a small number of markers.

Variation between patients is high. Tumours assigned the same anatomical stage can show very different gene expression profiles, and some early-stage and late-stage tumours appear biologically similar. This reflects tumour heterogeneity, immune influences, and background technical variation present in large public datasets. These sources of variation place a natural limit on how accurately cancer stage can be inferred from gene expression alone.

Cancer stage is defined using anatomical criteria, while gene expression reflects internal cellular behaviour at a specific point in time. For this reason, perfect prediction of cancer stage from gene expression alone is not biologically realistic (Kourou et al., 2015).

Some studies report moderate discrimination between stages under these biological limits (Miller et al., 2024). Performance above a simple baseline therefore reflects the presence of real biological signal, even though that signal is expected to be limited in strength.

3.3 Machine Learning Models in Cancer Research

Machine learning models used in cancer research were reviewed with the specific modelling choices in this project in mind rather than as a broad survey of the field. The focus was on studies that apply machine learning to gene expression data to predict clinical labels, which closely matches the structure of this project. Across the literature, linear models, tree-based models, and ensemble methods appear most often. This aligns with the four models selected here: logistic regression, linear support vector machines, random forests, and XGBoost.

Many gene expression studies use datasets containing hundreds to a few thousand patients and tens of thousands of genes. The dataset used in this project contained approximately 1,192 patients and around 20,530 genes before filtering, which is typical for this type of analysis. This matters because model behaviour is strongly influenced by the relationship between sample size and feature count. When the number of features greatly exceeds the number of samples, models are at high risk of learning unstable or noisy patterns rather than relationships that hold more broadly.

Across the literature, reported performance in comparable studies often falls between approximately 0.60 and 0.75 for accuracy or ROC-AUC when predicting clinical outcomes from gene expression alone. This range reflects the difficulty of learning from high-dimensional data where relevant signal is weak and distributed. Higher values are more often reported when prediction tasks are simpler, labels are more direct, or evaluation procedures are less strict.

Overfitting is therefore a recurring concern in gene expression studies. Overfitting occurs when a model captures patterns specific to the training data rather than relationships that generalise to new samples. This risk is especially high when feature selection and model training are not carefully controlled, given the large imbalance between the number of genes and the number of patients.

These considerations informed the modelling choices made in this project. I used logistic regression and linear support vector machines because they impose strong limits that reduce sensitivity to noise. Random forests and XGBoost were included to test whether modelling non-linear relationships could extract additional signal while still controlling complexity. Feature selection was applied before model training to reduce dimensionality and limit memorisation of spurious patterns.

Across all four models, performance values were similar and reached a clear plateau. This pattern is consistent with limits imposed by the structure of the data rather than by the choice of machine learning algorithm.

3.4 What Previous Studies Leave Unanswered

The literature shows that many machine learning studies using gene expression struggle to transfer to new patient groups. Models often perform well when evaluated on the same dataset used for training, but performance frequently drops when applied to independent cohorts. This pattern suggests that models commonly learn dataset-specific structure rather than biological signal that holds across populations (Vabalas et al., 2019).

A related issue is the limited use of true external testing. Reviews of machine learning applications in cancer research report that many studies do not evaluate models on independent datasets generated using different laboratory methods or drawn from different patient populations. Without such testing, it is difficult to determine whether reported performance reflects transferable biology or effects introduced by data collection, preprocessing, or cohort-specific bias. This gap directly informed the decision to evaluate model behaviour on the SCAN-B cohort rather than relying only on results from TCGA.

The literature also highlights recurring methodological weaknesses. Many studies report high accuracy or ROC-AUC values without fully addressing risks such as class imbalance, data leakage, or poor separation between training and evaluation steps. Earlier methodological work shows that practices such as performing feature selection before cross-validation can produce biased and overly optimistic results that fail to transfer to new data (Varma & Simon, 2006; Pudova et al., 2025). Similar concerns continue to appear in more recent reviews, particularly in studies using small datasets with very large numbers of features. These issues shaped the evaluation strategy used in this project and motivated cautious validation and comparison with simple baselines.

Several studies also note that performance limits in gene expression-based prediction tasks arise from the structure of the data and evaluation practices rather than from a complete absence of signal. The imbalance between sample count and feature count makes stable learning difficult, and more complex models usually require much larger datasets to

generalise reliably. While combining gene expression with additional molecular data can improve robustness, such data were not available for this project.

Taken together, the literature leaves an unresolved question: whether stage-related biological signal can be detected reliably when gene expression is evaluated under strict conditions that include exclusion of anatomical variables, careful control of evaluation steps, and testing on independent cohorts. Under these conditions, models are expected to achieve moderate but stable performance, which reflects the presence of real biological information while also defining the boundary of what gene expression alone can support.

Chapter 4. Methodology

4.1 Overall Project Design

This project was designed around a binary early versus late-stage classification because this boundary is both clinically meaningful and realistically learnable from gene expression data. Stages I to IIB were grouped as early-stage disease, and stages III to IV were grouped as late-stage disease. This reflects clinical practice, where the shift from early to late stage often corresponds to a clear change in treatment urgency and management approach.

This framing also improved label stability. Fine-grained stage categories are often unevenly represented and inconsistently annotated in public datasets. Attempting multi-class staging would have increased label noise and reduced the chance of learning patterns driven by biological differences rather than annotation effects. Focusing on a single, clinically important boundary improved interpretability and reduced ambiguity during training and evaluation.

Breast cancer was selected because it offers one of the most comprehensive publicly available gene expression resources with matched clinical stage information. I chose the TCGA Breast Invasive Carcinoma cohort because it provides RNA-seq data and carefully curated stage labels at a scale suitable for controlled analysis. Breast cancer is also biologically heterogeneous, with substantial variation between tumours across patients. This heterogeneity creates a demanding test case in which any consistent signal is unlikely to arise from an overly simplified biological setting.

Several design choices were shaped by practical limits related to dataset size, computing resources, and project timeframe. Methods requiring very large cohorts, highly complex architectures, or specialised clinical annotation were avoided. The analysis was restricted to bulk gene expression rather than combining multiple molecular layers. While multi-omics approaches can improve performance, they also increase

complexity and reduce interpretability. Keeping the scope tight reduced overfitting risk and ensured that each modelling decision could be clearly justified and repeated.

The design also reflects real hospital workflows. Useful tools must operate reliably under limited conditions, use data that already exists, and produce results quickly.

The methodology therefore prioritised stability, transparency, and realistic deployment rather than maximising performance under ideal conditions.

4.2 Dataset Selection and Structure

The TCGA Breast Invasive Carcinoma cohort was selected as the main dataset because it provides paired RNA-seq gene expression data and clinically curated stage labels within a single, well-documented resource. The dataset included approximately 1,192 patients with available stage information. Each sample contained expression measurements for around 20,530 genes generated using Illumina HiSeq sequencing. Log2-transformed, RSEM-normalised expression values provided by TCGA were used, as these reduce technical variation related to sequencing depth and library size and allow fair comparison across samples.

Stage labels were structured by grouping stages I, IA, IB, II, IIA, and IIB as early-stage disease, and stages III, IIIA, IIIB, IIIC, and IV as late-stage disease. Preserving finer distinctions would have increased label noise due to uneven representation and inconsistent annotation. A binary grouping improved reliability while keeping a distinction that directly affects clinical decision-making.

A clear class imbalance was present in the TCGA cohort, with early-stage samples greatly outnumbering late-stage samples. This imbalance was treated as a natural feature of real staging data rather than a problem to be artificially corrected. I avoided aggressive rebalancing because it would have distorted the underlying distribution and risked inflating performance estimates. Instead, imbalance was handled during evaluation by comparison with a majority-class baseline and by focusing on discrimination-based metrics alongside raw accuracy.

Generalisation was assessed using the SCAN-B cohort as a fully independent dataset. SCAN-B contains approximately 3,400 breast cancer samples generated using different laboratory methods and represents a separate patient population. I used this cohort to test whether gene panels selected from TCGA could be applied without modification. This choice prioritised stable behaviour over optimisation and reflects how models must perform when transferred between institutions.

Samples with missing or unclear stage labels were removed to prevent mislabelling from affecting training and evaluation. Because stage labels define the prediction target, keeping uncertain samples would have reduced the validity of performance estimates.

Dimensionality was reduced by applying variance filtering to remove genes with little variation across samples. This reduced the feature space from approximately 20,530 genes to around 4,000 genes. Low-variance genes add complexity without contributing useful discriminatory information, especially when the number of features greatly exceeds the number of samples.

Stability selection was then applied to further refine the feature set. Feature selection was repeated across 50 training subsamples, and genes were ranked based on how consistently they were selected. This approach favours repeatable signal over chance correlations and reduces sensitivity to individual train–test splits.

Finally, feature sets were restricted to compact gene panels of approximately 30 to 50 genes. Small panels reduce overfitting risk, improve interpretability, and match realistic laboratory limits. Larger panels would have increased model flexibility at the cost of stability and encouraged memorisation of dataset-specific patterns rather than reliance on robust biological signal.

4.3 Feature Selection and Dimensionality Reduction

Feature selection was treated as a central part of the methodology rather than a minor preprocessing step. I first applied variance filtering to reduce the feature space from approximately 20,530 genes to around 4,000 genes. Genes with very low variance across samples were removed because they are unlikely to contribute useful information for distinguishing between early-stage and late-stage disease. Keeping these genes would have increased complexity without adding meaningful signal.

Stability selection was then applied to identify genes that were consistently informative across different subsets of the training data. This process was carried out using 50 subsamples drawn from the training set. For each subsample, feature selection was performed independently and the selected genes were recorded. Genes that appeared repeatedly across subsamples were retained, while genes selected infrequently were treated as noise rather than repeatable biological signal.

This procedure reduced dependence on any single data split and improved reproducibility. Without stability selection, feature sets varied widely between runs, which would have made interpretation and external testing unreliable.

Final feature sets were restricted to compact gene panels of approximately 30 to 50 genes. I fixed panel size before evaluation on the held-out test set and based it only on selection frequency within the training data. This prevented indirect tuning on the test set and ensured that reported performance reflected genuine generalisation. Small panels also improve interpretability and fit practical deployment limits.

The same feature selection pipeline was applied across all models to ensure fair comparison. Feature selection was carried out using training data only and kept fully separate from the held-out test set. This separation was required to prevent information leakage and to ensure that evaluation reflected genuine model behaviour rather than methodological effects.

4.4 Model Selection and Training Process

Four machine learning models spanning a range of complexity and assumptions were selected: logistic regression, linear support vector machines, random forests, and XGBoost. The aim was not to identify a single best-performing model but to test whether increasing model flexibility improves the extraction of stage-related information from gene expression data. Using multiple models under the same preprocessing and feature selection pipeline allowed differences in behaviour to be linked to model structure rather than differences in data handling.

Logistic regression was used as a constrained linear baseline. Strong regularisation makes it suitable for high-dimensional settings where overfitting is a major risk. This model tested whether stage-related information could be captured through weighted combinations of gene expression values alone, without introducing complex interactions. It also provided a clear reference point for interpreting the behaviour of more flexible models.

A linear support vector machine was included because it performs well when the number of features exceeds the number of samples, which is common for gene expression data. Margin-based optimisation helps control complexity and reduce sensitivity to noise. Linear SVMs are widely used in genomic research and allowed comparison between two linear approaches with different optimisation strategies.

Random forests were included to test whether non-linear interactions between genes contributed additional information. Random forests can capture interactions without explicit feature engineering, but they are prone to overfitting in high-dimensional settings. To limit this risk, I constrained tree depth and restricted the number of features considered at each split. These controls reduced memorisation of training data and encouraged learning of consistent structure rather than noise.

XGBoost was included as a more flexible non-linear model. I used it to test whether small gains could be achieved beyond random forests under careful regularisation. Learning rate, tree depth, and the number of

boosting rounds were constrained to limit model complexity. These controls were needed because boosted tree models can fit spurious patterns and produce misleadingly high performance estimates if left unconstrained.

All models were trained using an 80/20 train–test split on the TCGA dataset. Within the training set, five-fold cross-validation was used for hyperparameter tuning and stability assessment. The test set was kept fully separate and used only for final evaluation. This separation ensured that reported performance reflected generalisation rather than information leakage or tuning effects.

This model selection strategy reflects real clinical limits, where systems must behave reliably with limited data, maintain stable performance across methods, and avoid reliance on a single highly tuned model.

4.5 Testing and Evaluation Strategy

Model evaluation used a strictly separated training and testing setup to prevent information leakage. The TCGA dataset was split into an 80% training set and a 20% held-out test set. I used the training data for feature selection, model fitting, and hyperparameter tuning, while the test set was kept only for final evaluation. This separation ensured that performance estimates reflected real learning rather than memorisation.

The SCAN-B dataset was used for external evaluation to assess generalisation behaviour and to report headline performance values. Gene panels selected from TCGA were mapped directly onto the SCAN-B gene expression data, and predictions were generated using models trained only on TCGA. No retraining or tuning was performed on SCAN-B. This preserved its role as an independent reference and allowed direct testing of whether learned patterns transferred across cohorts generated under different experimental conditions.

Performance on the TCGA test set was evaluated using accuracy and the area under the receiver operating characteristic curve (ROC-AUC). Accuracy measured the proportion of correct predictions at a fixed decision threshold, while ROC-AUC measured class separation across all thresholds. Both metrics were used because accuracy alone can be misleading in imbalanced datasets, whereas ROC-AUC provides a more reliable measure of discrimination.

This evaluation framework supports cautious interpretation of results. Strict data separation, comparison with simple baselines, and independent external testing allow distinction between dataset-specific fitting and behaviour that reflects real but limited biological signal. Under this framework, gene expression is evaluated as an early supportive signal within existing clinical workflows rather than as a standalone staging tool.

Chapter 5. Results

5.1 Effect of Gene Panel Size on Model Performance

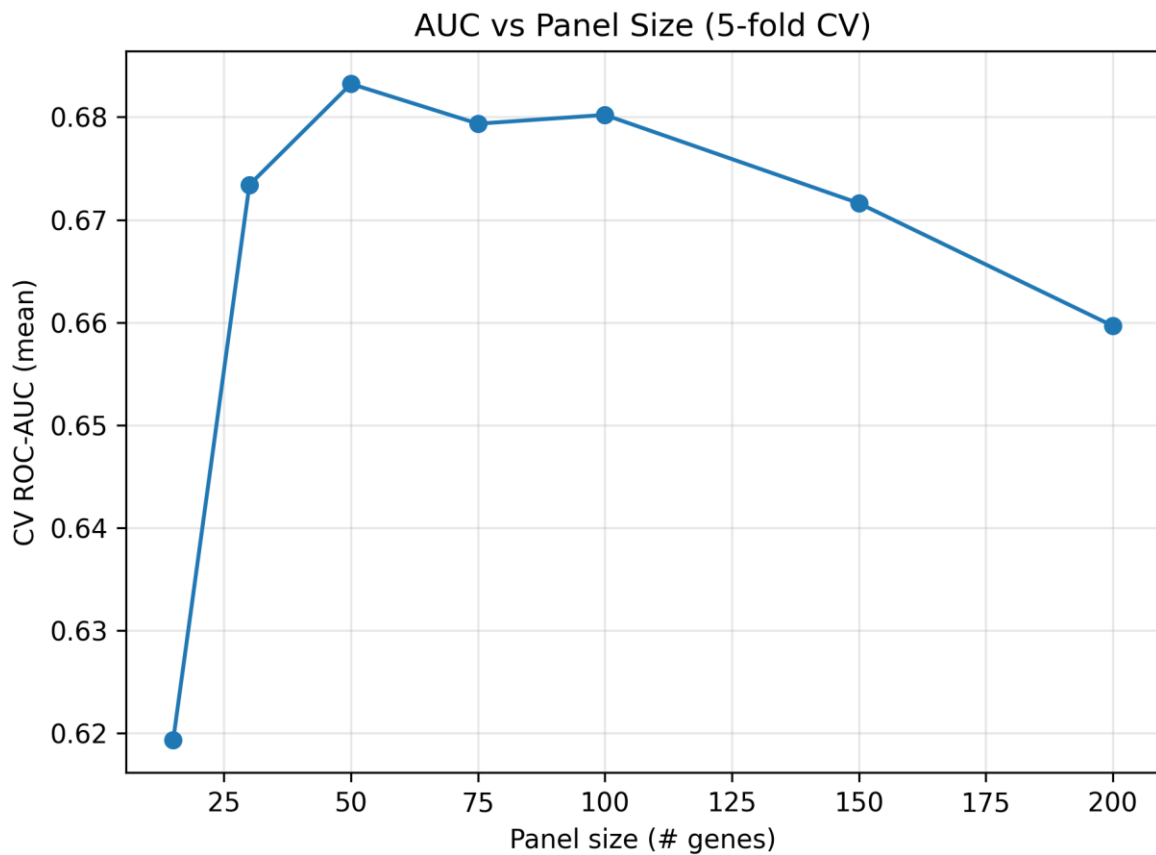


Figure 5.1 shows how model performance changes as the number of genes included in the input panel varies. The horizontal axis shows the number of genes used, and the vertical axis shows ROC-AUC obtained through cross-validation.

Very small gene panels produce low performance. With few genes, the input does not contain enough biological information to reliably distinguish between early-stage and late-stage disease. As panel size increases, performance improves sharply, showing that additional genes contribute stage-related signal.

Performance reaches a maximum at approximately 50 genes. At this panel size, most of the stage-related information present in the gene expression

data is captured. When additional genes are added beyond this point, ROC-AUC decreases as panel size continues to increase.

This pattern shows that larger panels introduce more noise than useful signal. Increasing the number of input genes beyond the optimal range reduces model stability rather than improving discrimination. A panel size of around 50 genes balances biological information content, robustness, and practical feasibility.

5.2 Behaviour on an Independent Dataset (SCAN-B)

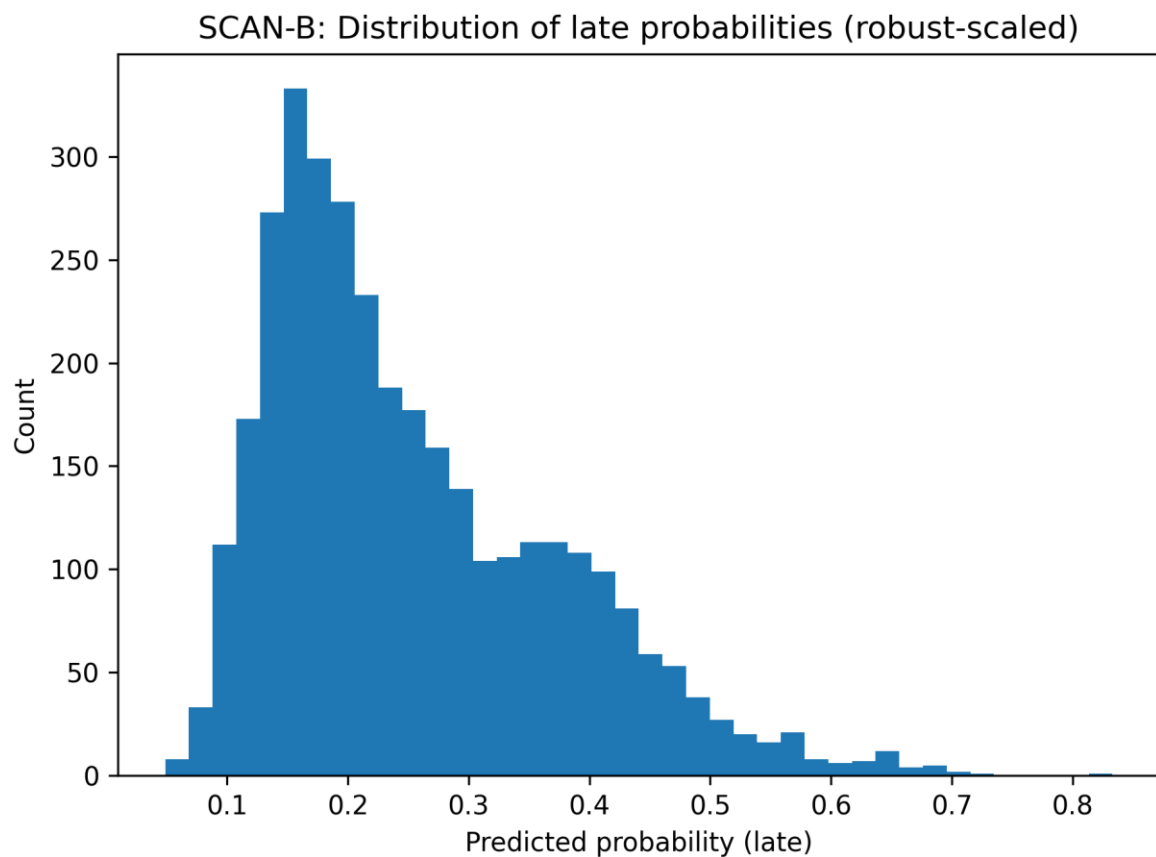


Figure 5.X shows model behaviour when applied to the independent SCAN-B breast cancer cohort, which was not used during training, feature selection, or hyperparameter tuning. This analysis examines how predicted probabilities are distributed across previously unseen patients.

The horizontal axis shows the predicted probability of late-stage disease, and the vertical axis shows the number of patients assigned to each probability range. Most patients receive low to medium predicted probabilities, indicating cautious model behaviour and an absence of extreme confidence.

A smaller fraction of patients are assigned higher late-stage probabilities, forming a narrow right-hand tail in the distribution. This pattern matches the known composition of the SCAN-B cohort, in which approximately one quarter of patients are late-stage. The remaining majority are assigned low probabilities and are classified as early-stage.

The model was applied without retraining or tuning. The resulting probability distribution reflects external evaluation rather than optimisation to the SCAN-B data. The shape and stability of the distribution indicate that the learned signal transfers beyond the TCGA dataset.

This functions as a transfer test rather than a performance benchmark, showing that predicted probabilities align with the expected prevalence of late-stage disease in an unseen cohort rather than collapsing toward extreme or dataset-specific behaviour.

5.3 ROC Curve Analysis Using Logistic Regression

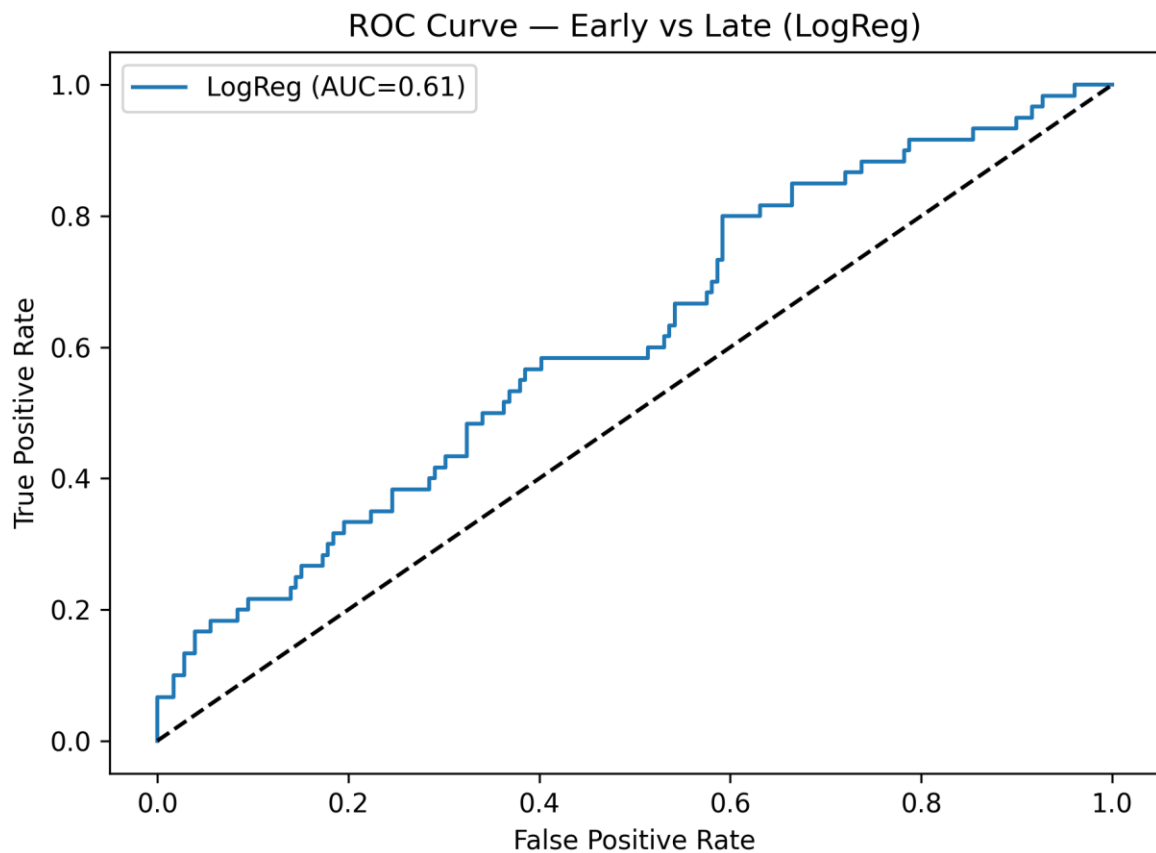


Figure 5.X shows the ROC curve for a logistic regression model trained using the selected gene panel and evaluated on the held-out test data. The dashed diagonal line represents random guessing.

The ROC curve lies above the diagonal across most thresholds, showing that the model ranks late-stage cases higher than early-stage cases more often than chance. The ROC-AUC is approximately 0.61, meaning that a randomly chosen late-stage patient is ranked higher than a randomly chosen early-stage patient about 61% of the time.

This performance is achieved using a constrained linear model that learns a single decision boundary and does not capture complex gene–gene interactions. The presence of measurable discrimination under these limits indicates that stage-related information is spread across many genes rather than driven by a small number of dominant markers.

5.4 ROC Curve Analysis Using XGBoost

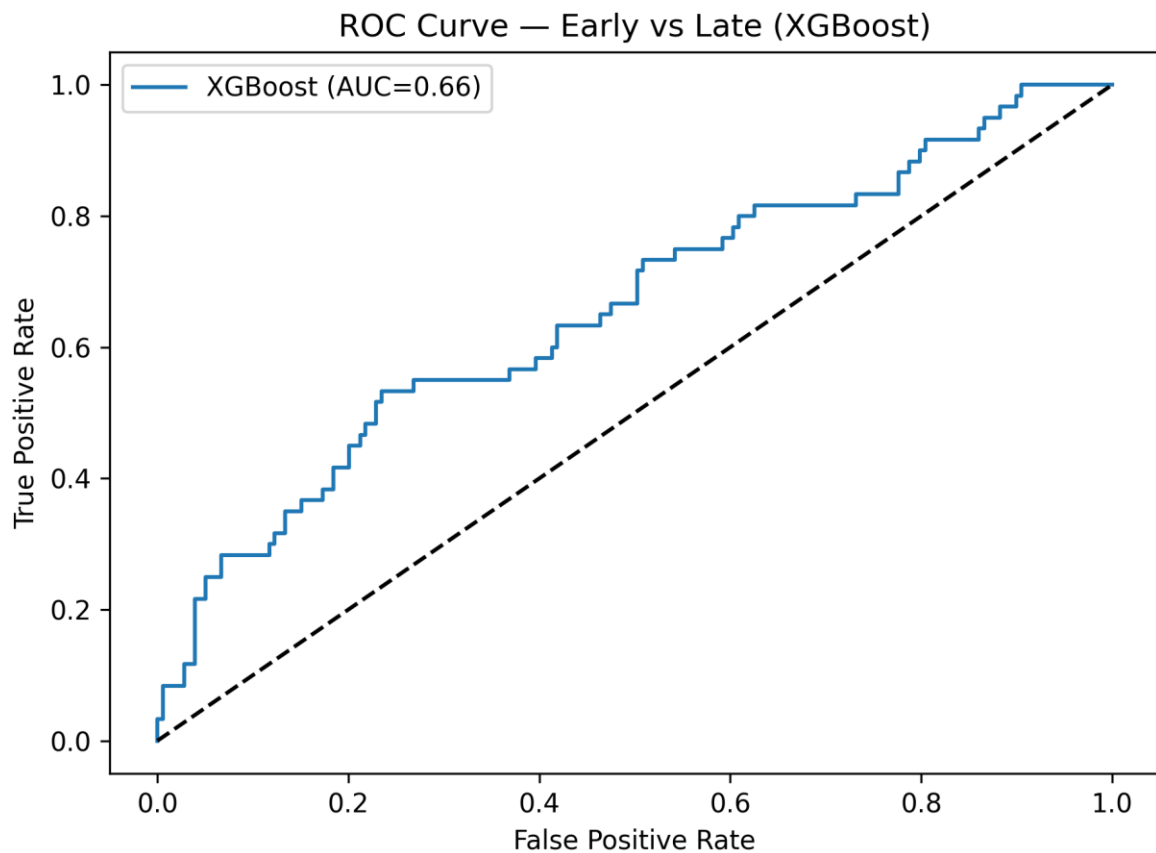


Figure 5.X shows the ROC curve for the XGBoost model trained on the same gene panel. The dashed diagonal line again represents random guessing.

The ROC curve remains above the diagonal across nearly the full range of thresholds. The ROC-AUC is approximately 0.66, indicating that when one early-stage and one late-stage patient are selected at random, the model correctly ranks the late-stage patient higher in around two thirds of cases.

Relative to linear models, XGBoost provides a modest improvement in discrimination. This difference indicates that most of the stage-related signal is captured by linear relationships, with a smaller contribution from non-linear interactions between genes. The limited size of the improvement shows that overall performance is constrained by the biological information present in the data rather than by model complexity.

5.5 Results Integrated Across Analyses

Model performance increases as gene panel size grows, peaks at approximately 50 genes, and declines when additional genes are added. This behaviour reflects a trade-off between biological signal and noise in high-dimensional gene expression data.

Across all evaluated models, performance values fall within a narrow range. Similar performance across model types indicates that discrimination is limited by the available biological signal rather than by algorithm choice.

When applied to the SCAN-B cohort without retraining or tuning, the model produces stable and realistic probability distributions. Most patients are assigned low to medium late-stage probabilities, with a smaller subset flagged at higher risk. This behaviour is consistent with the known stage distribution and shows transfer of the learned signal beyond the training dataset.

Table 5.1: Model performance on the TCGA held-out test set

Model	Accuracy	ROC-AUC
Logistic Regression	≈ 0.68	≈ 0.61
Linear SVM	≈ 0.69	≈ 0.65
Random Forest	≈ 0.68	≈ 0.658
XGBoost	≈ 0.69	≈ 0.66

Among the evaluated models, XGBoost produced the highest ROC-AUC under identical training conditions. This suggests that its ability to combine weak, non-linear signals across multiple genes allowed it to extract slightly more discriminative information than the linear models. At the same time, the small performance differences across models indicate that overall performance was mainly constrained by the strength of the underlying biological signal rather than by model choice.

In this setting, moderate and stable performance is not a weakness of the approach but evidence that a real biological signal has been isolated under strict constraints.

Chapter 6. Discussion

6.1 What My Results Actually Show

Gene expression data contains measurable but limited information about cancer stage at a population level. Across all evaluated models, accuracy values were approximately 0.68 when distinguishing between early-stage and late-stage disease. Because the dataset is imbalanced, accuracy is reported mainly for intuition, while discrimination-based metrics such as ROC-AUC better reflect model behaviour.

This level of performance is meaningful because the prediction task is indirect. Cancer stage is defined using anatomical features such as tumour size and spread, whereas gene expression reflects biological activity within tumour cells. These perspectives describe different aspects of disease, so any relationship between them is expected to be noisy rather than exact.

Model performance was compared with a majority-class baseline that predicts all samples as early-stage disease. This baseline achieves relatively high accuracy because early-stage cases are more common in the dataset. All models exceeded this baseline, showing that predictions relied on structure in the data rather than class frequency alone. The improvement indicates that models responded to consistent differences in gene expression across groups of samples, even though those differences were weak.

Performance was similar across model types. Increasing model flexibility from linear models to tree-based and boosted models did not produce sustained gains, indicating that non-linear structure could not extract additional stage-related signal. Expanding gene panels beyond the optimal range also reduced performance rather than improving it, showing that additional features introduced noise rather than hidden information.

Logistic regression, linear support vector machines, random forests, and XGBoost all produced comparable accuracy and ROC-AUC values. This convergence shows that discrimination was limited by the amount of

stage-related biological information present in the data rather than by model choice. If overfitting or information leakage were driving results, more flexible models would be expected to show a clear advantage. Instead, all models reached the same performance plateau, indicating extraction of the same limited signal.

Generalisation was assessed by applying trained models to the independent SCAN-B cohort. The emphasis was placed on prediction stability rather than headline accuracy. Stability refers to whether models produced coherent and consistent probability distributions when applied to new data using the same gene panels and trained parameters. Stable behaviour across cohorts indicates learning of transferable biological relationships rather than dataset-specific effects.

Performance above baseline reflects genuine biological signal rather than chance or class imbalance, while the observed plateau marks the limit imposed by tumour heterogeneity and the indirect nature of the prediction task.

6.2 Why Performance Is Limited

The observed limits in performance arise from the structure of the data and the nature of the problem rather than from shortcomings in model selection or implementation. The primary constraint is biological heterogeneity.

This heterogeneity arises from several biologically distinct sources. First, tumours assigned the same anatomical stage can differ widely in genetic drivers, growth behaviour, and cellular activity, weakening alignment between stage labels and gene expression profiles. Second, bulk gene expression reflects not only tumour cells but also immune infiltration and surrounding stromal tissue, adding variation related to the tumour environment rather than anatomical spread. Third, molecular subtype differences cut across stage boundaries, meaning that subtype-related expression patterns can dominate or mask weaker progression-related signals.

Dataset size further restricts performance. After filtering, the TCGA cohort contained approximately 1,192 samples, while each sample initially included measurements for more than 20,000 genes. Even after dimensionality reduction, the number of features remained large relative to the number of samples. In this setting, models can capture broad trends but struggle to learn subtle patterns that transfer reliably. Increasing model complexity mainly raises overfitting risk rather than improving real-world performance.

Public datasets introduce additional sources of noise. Cancer stage labels are assigned clinically and can vary slightly between institutions and over time. Although gene expression data is processed using standard pipelines, technical variation remains due to differences in sample handling and sequencing conditions.

Cancer stage is also an indirect target for gene expression-based prediction. Stage reflects anatomical spread, while gene expression reflects cellular behaviour at a specific point in time. These perspectives overlap but do not fully align. Some tumours show aggressive molecular

behaviour while remaining anatomically early-stage, while others are late-stage with more modest expression differences. This mismatch places a clear limit on predictive accuracy.

Under these conditions, moderate performance is an expected and informative outcome. Further gains would require additional data types, such as imaging or other molecular layers, rather than increased model complexity or alternative algorithms.

6.3 Comparison with the Literature

Findings were compared with published studies that apply machine learning to gene expression data for cancer-related classification tasks, with attention to work using similar data types, dataset sizes, and evaluation practices. Across this literature, moderate performance is reported consistently when predicting complex clinical labels from bulk RNA-seq data, especially when labels are indirect or biologically variable. Accuracy values around 0.68 and ROC-AUC values between approximately 0.65 and 0.70 fall within the range commonly reported.

Evaluation approach strongly influences reported performance. Best-practice guidelines stress strict separation between training and testing data and warn against methods that inflate accuracy, such as allowing feature selection to affect evaluation. Many studies reporting higher performance rely mainly on internal cross-validation or relaxed validation approaches, particularly in high-dimensional settings where results are highly sensitive to information leakage.

Published work also shows that increasing model complexity does not reliably improve transfer to new data. Highly flexible models can fit gene expression data well under controlled conditions, but this does not guarantee stable behaviour on independent datasets. Without external testing, high reported accuracy often reflects dataset-specific structure rather than biological signal that holds across populations.

These observations are consistent with reports that breast tumours show substantial molecular variation even within the same clinical stage. Gene expression studies repeatedly identify meaningful biological patterns, but not sharply separated boundaries for complex labels such as cancer stage. The results presented here follow this pattern and support careful, evidence-based interpretation.

6.4 Contribution of This Project

Gene expression data supports a measurable but limited role in cancer staging when evaluated under controlled conditions. Using gene expression alone, machine learning models distinguished between early-stage and late-stage breast cancer and consistently outperformed a simple majority-class baseline. These results clarify what stage-related information is present at a population level when anatomical and clinical variables are removed.

External testing was treated as a central design choice. Trained models were applied without modification to an independent cohort collected under different laboratory conditions, and evaluation focused on prediction behaviour rather than headline accuracy. This approach tests whether selected gene panels capture transferable biological patterns rather than dataset-specific effects and addresses a common gap in earlier work that relies mainly on internal validation.

Methodological transparency was prioritised throughout the project. Data filtering, feature selection, model constraints, and evaluation steps were documented in detail. Moderate performance was interpreted as evidence of real but limited biological signal rather than as readiness for clinical use. This interpretation follows accepted best practice in biomedical machine learning and avoids overstated conclusions.

Rather than aiming to maximise accuracy, this project set out to identify the boundary of what biology alone can realistically contribute to cancer staging. This question is rarely addressed directly in student or academic machine learning studies, where emphasis is often placed on improving performance without isolating biological limits.

Most machine learning pipelines focus on achieving the highest possible accuracy. In contrast, this project aimed to determine the maximum achievable signal under biologically realistic constraints, placing performance in context rather than treating it as an end in itself.

6.5 Handheld Demonstration Device and Computational Feasibility

This section addresses engineering feasibility rather than biological validity or model performance.

A small handheld demonstration device based on an ESP32 microcontroller was built to test whether computational limits contribute meaningfully to the observed performance boundaries. Its purpose was to show that, once a model is trained, inference on gene expression data does not require cloud computing or high-performance hardware.

The trained model and required preprocessing steps were implemented directly on the device. Gene expression values were transferred via USB, and all scaling and inference were performed locally. The device outputs predicted probabilities for early-stage versus late-stage disease rather than binary decisions. For demonstration purposes, publicly available and de-identified TCGA gene expression data prepared in advance were used to ensure stable and interpretable behaviour.

This device was built as a demonstration and educational tool rather than a clinical prototype. It illustrates the full workflow from gene expression input to probabilistic output and supports explanation of cautious interpretation, including cases where molecular predictions do not align with clinical stage. No claims are made regarding diagnostic use, clinical accuracy, or patient-level decision-making.

From a computational perspective, local inference on compact, low-power hardware is sufficient once a model has been trained. This shows that inference can be carried out under strict hardware limits, indicating that the main limits observed in this project arise from biological signal and data structure rather than from computational or engineering capacity. Further technical details are provided in the Appendices.

6.6 Project Novelty and Practical Significance

Cancer stage is defined anatomically using tumour size, lymph node involvement, and the presence of metastasis. These features are not directly encoded in gene expression data. From both a biological and clinical perspective, strong standalone prediction of stage from gene expression alone is therefore not expected. This project does not attempt to challenge that reality. Instead, it focuses on when gene expression data becomes available within real hospital workflows.

The central significance of this work is that gene expression data is often generated earlier than full anatomical staging. Imaging, pathology review, and multidisciplinary discussion usually occur in sequence and introduce substantial delays. This project shows that gene expression already contains a consistent and repeatable signal related to disease progression at this early point in the diagnostic timeline. By isolating gene expression from all anatomical and clinical staging inputs, the work measures how much stage-related information is available before scans and pathology are complete.

The contribution of this project is therefore not improved classification accuracy but workflow acceleration. Gene expression is positioned as an early molecular triage signal that operates before traditional staging steps.

Once molecular data is available, samples whose expression patterns align more closely with later-stage disease can be flagged immediately. This allows earlier prioritisation of imaging, pathology review, and multidisciplinary discussion. Decision-making is moved forward in time without changing clinical definitions, authority, or judgement.

A key operational advance is the compression of this signal into small and stable gene panels. Reducing approximately twenty thousand genes to panels of thirty to fifty genes transforms gene expression from a research-scale input into something compatible with real laboratory timelines. Small panels are faster to process, easier to standardise, and simpler to integrate into existing pipelines. This makes early molecular triage practical rather than theoretical.

The project deliberately prioritised generalisation and stability over internal optimisation. Similar performance across multiple model types and stable behaviour on an independent cohort generated under different laboratory methods indicate that the extracted signal is not specific to a single dataset. This matters in practice because a workflow support tool must behave predictably across hospitals, not just within one dataset.

The intended use of the system is limited and explicit. The models are not accurate enough to replace clinical staging, and no such use is proposed. Their value lies in time. By identifying cases more likely to correspond to later-stage disease as soon as gene expression data becomes available, hospitals can allocate attention and resources earlier. This reduces downstream delays while preserving existing clinical judgement.

Finally, inference on low-power embedded hardware shows that computation is not the limiting factor for deployment. Once trained, the model can generate predictions quickly and at low cost. This indicates that the main constraints on real-world use are organisational rather than technical. Data availability, clinical validation, and integration into existing workflows matter more than hardware capability. From a systems perspective, this reinforces the core conclusion of the project: biology provides the signal early, and current technology is already fast enough to act on it.

Chapter 7. Conclusion

This project examined whether machine learning models can distinguish between early-stage and late-stage breast cancer using gene expression data alone. The task was defined narrowly and deliberately. Imaging, tumour size, lymph node involvement, and other clinical variables used to assign stage were excluded so that any detected signal arose from biological differences rather than from staging definitions themselves. Framing the task as a binary early versus late-stage classification reduced label uncertainty and focused the analysis on what gene expression can realistically support.

Across all tested models, performance fell within a narrow and consistent range. Accuracy values were approximately 0.68, and ROC-AUC values ranged between 0.61 and 0.66. Increasing model complexity did not lead to meaningful improvements, and all models exceeded a majority-class baseline. Similar performance across model types indicates that the same underlying signal was being extracted and that discrimination was limited by available biological information rather than by algorithm choice or overfitting.

The results show that disease progression leaves a measurable molecular footprint in gene expression at a population level. At the same time, tumour heterogeneity, indirect label definitions, and dataset structure place a clear upper limit on what gene expression alone can achieve for cancer staging. This limit reflects biological and data-related constraints rather than weaknesses in model design or evaluation.

Once gene expression data is available, samples whose molecular patterns align more closely with later-stage disease can be flagged. This supports earlier prioritisation of imaging, pathology review, and multidisciplinary discussion within existing clinical workflows, reducing delays without changing established clinical authority or staging criteria. By enabling earlier molecular triage, this approach shortens the decision-making window before full anatomical staging is complete. In real clinical workflows, this earlier prioritisation has the potential to speed up imaging,

pathology review, and treatment planning without altering diagnostic rules or responsibility.

By identifying this limit rather than attempting to push beyond it, the project demonstrates realistic biological interpretation, which is essential for responsible use of machine learning in medicine.

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Data Sources

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Appendices

Appendix A: Final 50 Gene Panel

- **SPDEF** – A transcription factor involved in epithelial cell differentiation and glandular tissue development.
- **TTC22** – A gene encoding a protein with tetratricopeptide repeats, likely involved in protein–protein interactions.
- **LAMC2** – Encodes a subunit of laminin, a key component of the extracellular matrix involved in cell adhesion and migration.
- **UBXN10** – Encodes a UBX domain-containing protein associated with protein degradation pathways.
- **MMRN1** – Encodes multimerin-1, a protein involved in platelet function and extracellular matrix interactions.
- **SVEP1** – Encodes an extracellular matrix protein involved in cell adhesion and vascular development.
- **INMT** – Encodes indolethylamine N-methyltransferase, an enzyme involved in small-molecule metabolism.
- **APOL4** – Member of the apolipoprotein L family, involved in lipid transport and cellular stress responses.
- **PGM5** – Encodes phosphoglucomutase-like protein involved in cytoskeletal organisation.
- **LRRC26** – Encodes a regulatory subunit of potassium ion channels, influencing membrane electrical activity.
- **LHX1** – A transcription factor involved in early developmental patterning and cell fate determination.
- **CXCL5** – A chemokine involved in immune cell recruitment and inflammatory signalling.
- **FERMT1** – Encodes kindlin-1, a protein involved in cell adhesion and integrin signalling.
- **KLF15** – A transcription factor involved in metabolic regulation and cellular differentiation.
- **BAIAP3** – Encodes a protein associated with vesicle trafficking and regulated secretion.
- **CAPN11** – Member of the calpain family of proteases involved in protein cleavage and cellular remodelling.
- **CCR6** – Encodes a chemokine receptor involved in immune cell migration.
- **C1QTNF7** – A gene encoding a secreted protein related to immune and metabolic regulation.
- **ADD2** – Encodes beta-adducin, a cytoskeletal protein involved in cell membrane stability.
- **APBA2** – Encodes a protein involved in intracellular signalling and vesicle transport.
- **F10** – Encodes coagulation factor X, a key enzyme in the blood clotting pathway.
- **ADRA2C** – Encodes an alpha-2 adrenergic receptor involved in neurotransmitter regulation.
- **ADH1C** – Encodes alcohol dehydrogenase 1C, involved in alcohol and aldehyde metabolism.
- **MMP13** – Encodes a matrix metalloproteinase involved in extracellular matrix breakdown.
- **ZNF385B** – Encodes a zinc-finger protein likely involved in RNA binding and gene regulation.
- **AFF2** – Encodes a transcriptional regulator involved in RNA processing.
- **TMC3** – Encodes a transmembrane protein with a currently unclear biological function.
- **DLEC1** – Encodes a protein implicated in cell cycle regulation and cellular growth control.
- **SLC24A3** – Encodes a sodium/potassium/calcium exchanger involved in ion transport.
- **MEOX2** – A transcription factor involved in mesenchymal cell differentiation and vascular development.
- **PODN** – Encodes podocan, an extracellular matrix protein involved in tissue organisation.
- **SLC7A2** – Encodes an amino acid transporter involved in cellular nutrient uptake.
- **NR2F1** – A nuclear receptor involved in transcriptional regulation and developmental processes.
- **SAMD11** – Encodes a protein with a SAM domain, likely involved in signalling pathways.
- **AOC3** – Encodes an amine oxidase involved in leukocyte trafficking and vascular function.
- **NFASC** – Encodes neurofascin, a cell adhesion molecule involved in cell–cell interactions.
- **ARNTL2** – Encodes a transcription factor related to circadian rhythm regulation.
- **SCN3B** – Encodes a subunit of voltage-gated sodium channels affecting electrical signalling.
- **TBX21** – A transcription factor involved in immune cell differentiation.
- **ALDH1A1** – Encodes an aldehyde dehydrogenase involved in detoxification and retinoic acid metabolism.
- **EGR1** – An immediate-early transcription factor activated by cellular stress and growth signals.
- **STEAP4** – Encodes a metalloredutase involved in cellular metabolism and inflammatory responses.
- **HSPB2** – Encodes a small heat-shock protein involved in cellular stress protection.
- **KANK4** – Encodes a cytoskeletal-associated protein involved in cell shape regulation.
- **BCL2L14** – Member of the BCL-2 protein family involved in regulation of programmed cell death.
- **BCL2A1** – Encodes an anti-apoptotic protein involved in cell survival pathways.
- **AP3B2** – Encodes a subunit of the adaptor protein complex involved in intracellular trafficking.
- **COL27A1** – Encodes a collagen protein involved in extracellular matrix structure.
- **FOLR2** – Encodes a folate receptor involved in vitamin uptake and cellular metabolism.
- **TMPRSS4** – Encodes a transmembrane serine protease involved in protein processing at the cell surface.

Appendix B : XGBoost Model Technical Specification

Task

- **Type:** Supervised binary classification
- **Target:** Cancer stage
 - Class 0: Early stage
 - Class 1: Late stage

Input feature space

- **Feature count:** 50
- **Feature type:** Continuous (gene expression values)
- **Feature identity:** Fixed gene panel

Gene panel

- Panel defined explicitly as a fixed ordered list of 50 gene symbols
- No dynamic feature selection at inference time
- Feature order must be preserved exactly

Preprocessing

- **Scaling method:** Robust scaling using TCGA cohort statistics
- **Center:** Per-gene median
- **Scale:** Per-gene interquartile range (IQR)

For each gene:

$$x'_g = \frac{x_g - \text{median}_g}{\text{IQR}_g}$$

- Medians and IQRs are fixed constants
- Scaling parameters derived from TCGA training data only

Model class

- **Category:** Parametric ML classifier
- **Learning paradigm:** Empirical risk minimisation
- **Output:** Continuous score in $[0, 1]$ (class probability)

Class imbalance handling

- **Observed late-stage prevalence:** ~ 0.25
- **Imbalance strategy:** Cost-sensitive learning
- **Positive class weight:**
 - `scale_pos_weight` ≈ 4.48
- No resampling, no synthetic balancing

Decision rule

- **Decision threshold strategy:** Prevalence-based
- **Selected threshold:**
 - `threshold_mode` = "prev"
 - Numeric value ≈ 0.321

Decision:

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq \tau \\ 0 & \text{if } p < \tau \end{cases}$$

- Alternative thresholds (Youden, F1) computed but not used
- Threshold fixed post-training

Training configuration

- **Random seed:** 42
- **Determinism:** Yes (given identical inputs and preprocessing)
- **Hyperparameter tuning:** Limited / constrained
- **Objective:** Stable generalisation, not peak optimisation

Validation protocol

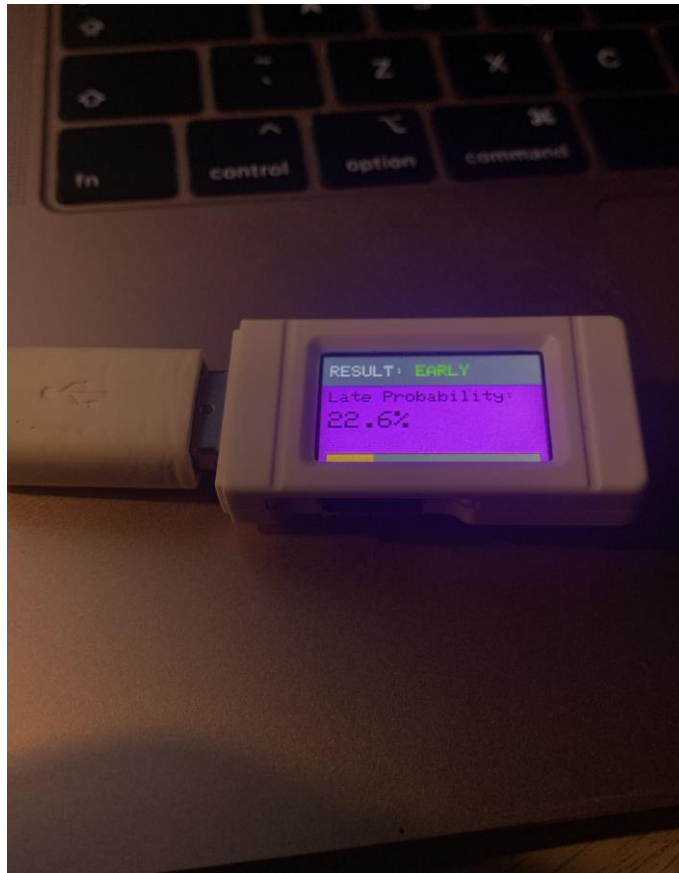
- Internal evaluation on held-out TCGA samples
- External evaluation on independent dataset
- No retraining or refitting on external data

Failure modes / constraints

- Assumes input genes match panel exactly
- Assumes expression units compatible with TCGA preprocessing
- Not calibrated beyond training distribution
- Population-level inference only

A cost-sensitive binary classifier operating on a fixed 50-gene, robust-scaled feature space, using prevalence-calibrated thresholding and deterministic training configuration.

Appendix C: ESP32-S3 Demonstration Device



Purpose of the Embedded Implementation

As part of this project, a custom embedded system was implemented to evaluate the trained stage classification model directly on hardware. The goal was not to retrain or optimise the model on-device, but to faithfully reproduce the inference pipeline developed during analysis in a constrained computing environment. This appendix documents how the trained model was translated into an embedded form and executed deterministically on an ESP32-S3 microcontroller.

Hardware as a Model Execution Target

The model is executed on a Waveshare ESP32-S3 board built around the ESP32-S3 system-on-chip from Espressif Systems. The hardware provides sufficient floating-point performance and memory to evaluate a reduced model while remaining fully standalone.

The ESP32-S3 is used strictly as a numerical inference engine and visualisation layer. No training, optimisation, or data-driven adaptation occurs on the device.

Model Representation on the Device

The original project model was trained using an ensemble-based method unsuitable for direct execution on microcontroller hardware. To enable embedded inference, the trained model was distilled into a compact parametric form that preserves the learned decision boundary while remaining computationally lightweight.

The embedded model consists of:

- A fixed vector of 50 learned weights
- A single bias term
- A binary decision threshold

All parameters are stored either:

- In a JSON file on the SD card, loaded at startup, or
- As embedded constants in firmware as a fallback

This representation ensures that the on-device model corresponds exactly to a fixed snapshot of the trained model.

Input Dimensionality and Gene Ordering

The embedded model operates on a fixed 50-gene panel, identical in size and ordering to that used during training and evaluation. The ESP32 expects a single input vector of 50 numerical expression values, provided in the correct order.

No gene selection, filtering, or reordering occurs on the device. Correct dimensionality and ordering are enforced explicitly to prevent invalid inference.

On-Device Normalisation Strategy

To ensure consistency with the training pipeline, each input value is normalised on-device using robust statistics computed from the TCGA training cohort. For each gene, the following values are stored:

- Median
- Interquartile range (IQR)

Normalisation is performed using:

$$\hat{x}_i = (x_i - \text{median}_i) / \text{IQR}_i$$

Safeguards are included to prevent division by zero or numerical instability. This step is critical: without identical normalisation, the embedded model would not be evaluating the same feature space as the trained model.

Inference Computation

After normalisation, the ESP32 computes a linear score:

$$z = \text{bias} + \sum (w_i \times \hat{x}_i)$$

The score is transformed using a sigmoid function to produce a probability value in the range [0, 1], representing the model's estimated likelihood of late-stage disease.

This computation mirrors the mathematical structure of the distilled model and is implemented using standard floating-point arithmetic.

Decision Thresholding

A fixed threshold, derived during model development, is applied to the output probability to generate a binary classification:

- EARLY if probability < threshold
- LATE if probability $\geq$ threshold

No adaptive thresholding or calibration occurs on-device. The threshold is treated as a model parameter and stored alongside the weights and preprocessing statistics.

Output Interpretation

The ESP32 presents:

- The binary stage classification

- The underlying probability value

Both values are direct outputs of the embedded model and are not post-processed beyond thresholding. The device therefore functions as a transparent inference endpoint rather than a heuristic or rule-based system.

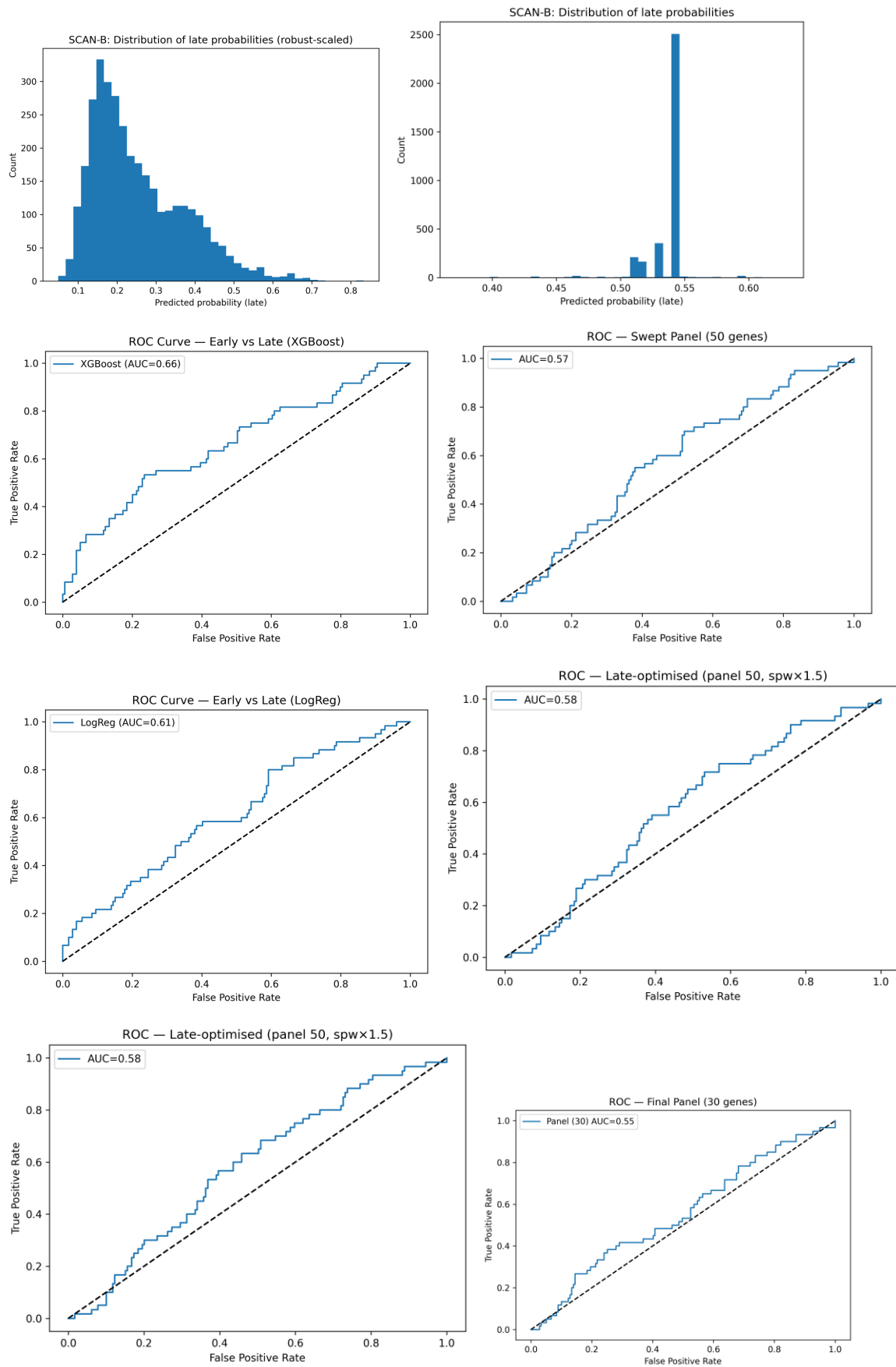
Scope and Limitations

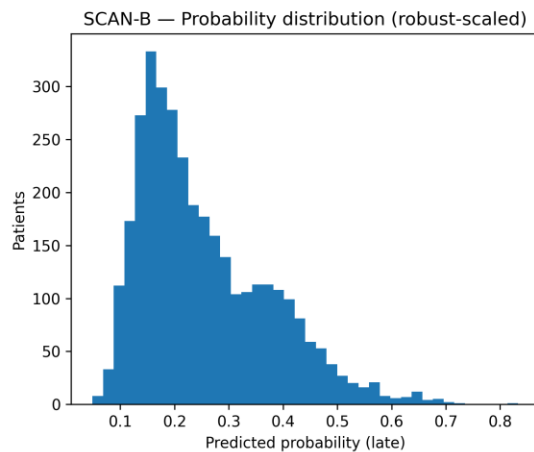
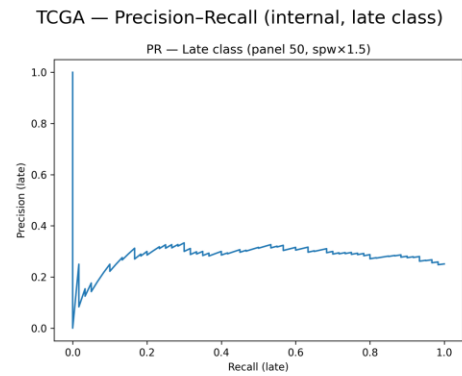
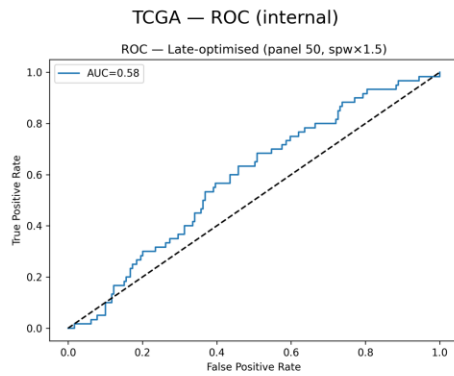
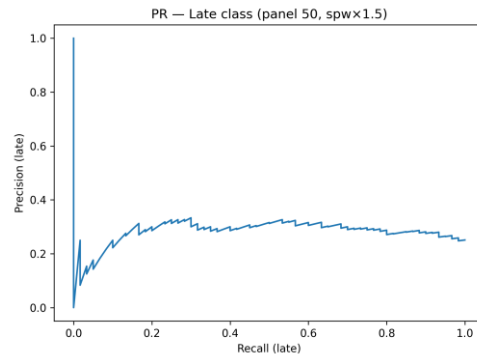
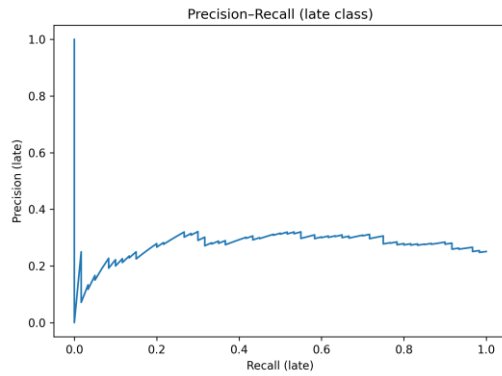
The embedded implementation is intentionally constrained:

- It evaluates a single fixed model
- It does not perform uncertainty estimation or confidence intervals
- It does not incorporate clinical metadata or imaging features
- It does not modify or retrain the model

Its purpose is to demonstrate that the signal learned during analysis can be evaluated reproducibly in a minimal, standalone computational environment.

Appendix D: Visuals





****Model & Panel****

- Panel size: 50 genes
- Example genes: SPDEF, TTC22, LAMC2, UBXN10, MMRN1, SVEP1 ...

****Thresholds (TCGA-derived)****

- Mode: prev
- Fixed threshold = 0.321
- Youden: 0.096
- F1(late): 0.048
- Prevalence-matched (TCGA): 0.321 (rate=0.251)

****External — SCAN-B (n≈3,409)****

- Fixed mode late proportion: 0.266
- SCAN-B matched late proportion: 0.251 (thr≈0.331)

Appendix E: Reproducibility and Data Integrity Summary

Datasets Used

Two publicly available breast cancer gene expression datasets were used:

- **TCGA Breast Invasive Carcinoma (TCGA-BRCA)**
RNA-sequencing gene expression data with curated clinical stage labels. This dataset served as the primary training and internal evaluation cohort.
- **SCAN-B (Sweden Cancerome Analysis Network – Breast, GSE96058)**
An independent international breast cancer cohort generated using different laboratory protocols and representing a separate patient population. This dataset was used exclusively for external evaluation.

No restricted or private patient data were accessed. All datasets are publicly available and fully de-identified.

Data Access and Preprocessing

Pre-processed gene expression values provided by the original data sources were used. Raw sequencing reads were not reprocessed as part of this project.

- TCGA expression values were log2-transformed and RSEM-normalised using standard pipelines provided by TCGA.
- SCAN-B expression values were used as provided by the dataset authors and mapped to the same gene identifiers where required.

Using standardised, widely accepted preprocessing pipelines reduced the risk that observed patterns arose from data handling rather than biological differences.

Feature Selection and Dimensionality Control

To address the high dimensionality of gene expression data, feature selection was treated as a core methodological step.

- Genes with very low variance across samples were removed using variance filtering.
- Stability selection was then applied using 50 subsamples drawn from the training data.
- Genes were ranked based on selection frequency across subsamples.
- A fixed gene panel of 50 genes was selected before final evaluation.

Feature selection was performed using training data only. No information from held-out test data or external datasets influenced gene selection.

Model Training and Evaluation Safeguards

To prevent information leakage and overly optimistic performance estimates, strict data separation was maintained throughout the analysis.

- The TCGA dataset was split into an 80% training set and a 20% held-out test set.
- Feature selection, model fitting, and hyperparameter tuning were performed using the training data only.
- The held-out test set was used exclusively for final internal evaluation.
- The SCAN-B dataset was used as a fully independent external cohort. No retraining, refitting, or threshold adjustment was performed on SCAN-B.

Performance was interpreted relative to a majority-class baseline to account for class imbalance. Discrimination-based metrics were reported alongside accuracy to avoid misleading conclusions.

Randomness Control and Determinism

Wherever possible, sources of randomness were controlled to ensure repeatability:

- A fixed random seed (42) was used for data splitting and model training.
- Model configurations were constrained and deterministic given identical inputs.
- Feature ordering and preprocessing parameters were fixed and documented.

This ensures that repeated runs under the same conditions produce equivalent results.